

Instruction manual



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Mars EdPal Instruments Pvt. Ltd.

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DEFLECTION OF TRUSS APPARATUS

INSTRUCTION MANUAL



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DEFLECTION OF TRUSS APPARATUS

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APPARATUS REQUIRED:

- | | | |
|-------------------|---|---------|
| • Dial gauge | : | 1 No. |
| • Magnetic Base | : | 1 No. |
| • Hangers | : | 3 Nos. |
| • Weight of 500gm | : | 10 Nos. |
| • Weight of 1 kg | : | 5 Nos. |

DEFLECTION OF TRUSS APPARATUS

OBJECTIVE:

Deflections of a truss – horizontal deflections and vertical deflections of various joint truss.

REQUIREMENTS:

1. Truss Apparatus
2. Weights.
3. Hangers

INTRODUCTION:

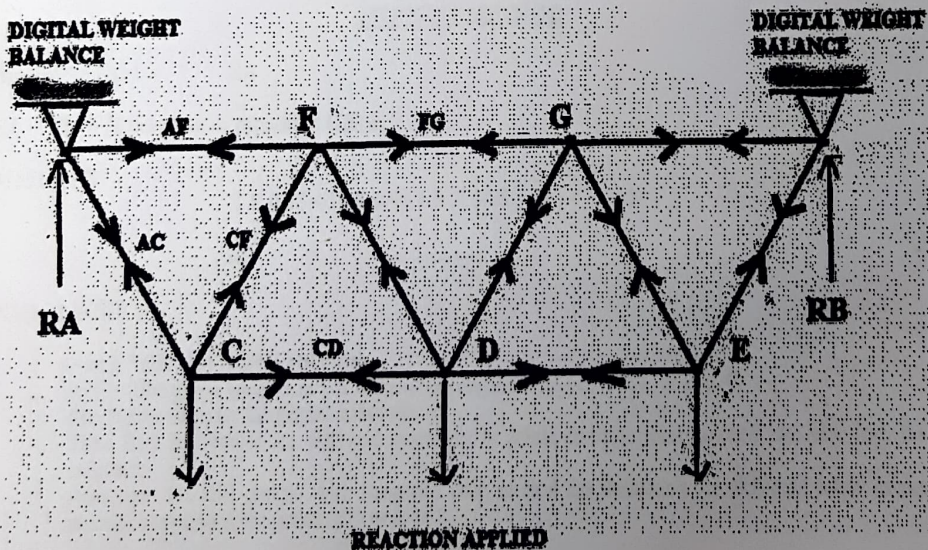
Trusses are structures composed of long thin rods fastened together at their ends. They are used because of their light weight, their ability to support a large variety of load systems, and because they are easily assembled. Trusses are easy to study in the laboratory, and when suitably idealized they are relatively simple to analyze. The goal of such analysis in practice is to determine whether a structure is safe for some planned or possible loading. In this lab you will learn some of the basic ideas about trusses by determining the forces in two welded or jointed truss structures, for a specific load and support systems, by two methods:

- a) experimentally, and
- b) using basic laws of static mechanics applied to an idealization of the structure.

THEORY:

The method of joints is one of the simplest methods for determining the force acting on the individual members of a truss because it only involves two force equilibrium equations. Since only two equations are involved, only two unknowns can be solved for at a time. Therefore, you need to solve the joints in a certain order. That is, you need to work from the sides towards the center of the truss. Since you need to work in a certain order, the Method of Sections (which will be covered later) can be more useful if you just want to know the forces acting on a particular member close to the center of the truss.

ΣF_y and ΣF_x method use at different point or different members



JOINT METHOD

Point RA	=	ΣF_x	=	$AF + AC \cos 60^\circ = 0$	
		ΣF_y	=	$AC \sin 60^\circ - RA$ (Spring Balance weight)	= 0
		AC	=		
		AF	=		
Point C	=	ΣF_y	=	$AC \sin 60^\circ + CF \sin 60^\circ - C$ (Reaction Applied)	= 0
		ΣF_x	=	$CF \cos 60^\circ + CD - AC \cos 60^\circ$	= 0
		CF	=		
		CD	=		
Point F	=	ΣF_y	=	$FD \sin 60^\circ + CF \sin 60^\circ$	= 0
		ΣF_x	=	$AF + FD \cos 60^\circ - CF \cos 60^\circ - FG$	= 0
		FD	=		
		FG	=		
Point D	=	ΣF_y	=	$FD \sin 60^\circ + DG \sin 60^\circ - D$ (Reaction Applied)	= 0
		ΣF_x	=	$DG \cos 60^\circ + DE - FD \cos 60^\circ - CD$	= 0
		DG	=		
		DE	=		
Point G	=	ΣF_y	=	$GD \sin 60^\circ + GE \sin 60^\circ$	= 0
		ΣF_x	=	$FG + DG \cos 60^\circ - GB - GE \cos 60^\circ$	= 0
		GE	=		
		GB	=		
Point RB	=	ΣF_y	=	RB (Spring Balance weight) - $BE \sin 60^\circ$	= 0
		BE	=		

MOMENT METHOD

Moment at Point = RA

RA + RB = Total reaction applied at C, D & E

Find the RB

$$C \times 0.15 + D \times 0.45 + E \times 0.75 = RB$$

$$RB = \dots\dots\dots$$

PROCEDURE :

1. Firstly "ON" the Digital Balances.
2. Applied the given weights as per experiment requirement point of C or D or E.
3. See the readings on the digital balances.
4. Then find the trusses on different members or joints through joint method and also find the moment.
5. Also used the dial gauges with magnetic base to find the deflection for different joints as per reactions applied at the point of C or D or E.
6. RA and RB is shown in the Digital Balances.

OBSERVATION:

S.NO	NODE DEFLECTION	POINT "C"	POINT "D"	POINT "E"
1.	Initial Dial Gauge Reading (mm)			
2.	Additional loads (Kg's)	1	2	1
3.	Final Dial Gauge Reading (mm)			
4.	Deflection (3) - (1) (mm)			

PRECAUTIONS:

1. Apply Loads without any jerk.
 2. Measure the deflection to the nearest of a millimeter.
 3. Perform the experiment at a location, which is away from any external disturbance.
 4. Ensure that the support are rigid.
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